

Dynamical systems time series classification

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We propose a method for time series classification under the assumption that series of different classes are generated by distinct dynamical systems, modeled as ordinary or stochastic differential equations whose noisy phase trajectories form the observed data. The classifier assigns each series to the class whose dynamics best fit it in a certain sense, and we prove it is asymptotically optimal under suitable noise conditions. Since exact trajectories are rarely available, we replace them with conditional expectations obtained by solving a smoothing problem, and learn unknown vector fields from data via Neural Ordinary Differential Equations. The method is validated on synthetic linear systems, human activity recognition from inertial sensors, and heart failure detection from ECG recordings.

1 Introduction

The paper introduces a method for time series classification. A *time series* is a finite sequence of scalar values $x(t_i) = \{x_{t_1}, \dots, x_{t_n}\}$ of variable length defined at particular

times $t_1, \dots, t_n \in \mathbb{R}$. A *multivariate* time series is a sequence of vector values $\mathbf{x}(t_i) = \{\mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}\}$, where $\mathbf{x}_{t_i} \in \mathbb{R}^d$. This is equivalent to having d time series with the same times. In what follows we assume time series to be multivariate in general. From a theoretical point of view it is convenient to assume the time series values $\mathbf{x}_{t_i}$ to be *random vectors*. Time series have an associated *class label* $y \in \{1, \dots, K\}$ that is also random in general. Here K is the number of classes. A *classifier* is a map $\delta(\mathbf{x}(t_i))$ that associates any time series with a class label. We make the notion of a “good” classifier precise in Section 2.

There are many methods for time series classification in the literature. *Distance-based* approaches rely on measuring similarity between time series using distance metrics. The metric can be defined using Dynamic Time Warping [1], and then the k-Nearest-Neighbors (k-NN) classifier can be used [2]. *Feature-based* approaches extract a finite set of statistical [3, 4] or spectral [5] features from time series for further classification. *Deep learning* approaches use neural networks to automatically learn time series features from raw training data. Convolutional neural networks [6, 7], recurrent neural networks (RNN) [8] or their combination [9–11] are designed to process sequential data like time series.

In the paper we assume a particular model behind the time series — a *dynamical system*. By that we mean the time-invariant *ordinary differential equation (ODE)* of the form

$$\begin{cases} \frac{d}{dt} \bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}}), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (1)$$

where $\bar{\mathbf{x}} \in M$, $f : M \rightarrow M$ is a *vector field*, $\bar{\mathbf{x}}_0$ is a *starting point*, and M is a differentiable manifold in general. We constrain the vector fields to belong to a normed space $f \in (\mathcal{F}(M), \|\cdot\|)$ with properties specified later. To find a solution $\bar{\mathbf{x}}(t)$ satisfying (1) on some time interval $t \in [0, T]$ is to solve an *initial value problem (IVP)*. We also denote the IVP solution as $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)$ and call it a *phase trajectory*. Under some

regularity conditions on f the solution is unique [12]. We associate a different vector field f_j , $j \in \{1, \dots, K\}$, with each series class. The series values are the noisy phase trajectories

$$\mathbf{x}(t_i) = \bar{\mathbf{x}}(t_i) + \boldsymbol{\epsilon}(t_i), \quad (2)$$

where $\boldsymbol{\epsilon}(t_i)$ is a random noise process. Its values $\boldsymbol{\epsilon}_{t_i}$ are i.i.d. random vectors (across all classes and times) with

$$\mathbb{E}[\boldsymbol{\epsilon}_{t_i}] = 0, \text{ Cov}(\boldsymbol{\epsilon}_{t_i}) = R \text{ where } \|R\|_2 \leq \sigma^2 < \infty. \quad (3)$$

Here R is the covariance matrix and $\|R\|_2$ is the matrix spectral norm. Therefore $\mathbb{E}[\mathbf{x}(t_i)] = \bar{\mathbf{x}}(t_i)$ and $\text{Cov}[\mathbf{x}(t_i)] = R$. In general, the classifier must account for both the deterministic component $\bar{\mathbf{x}}(t_i)$ and the noise component $\boldsymbol{\epsilon}(t_i)$ of the series. We will later show that under some conditions there is an asymptotically accurate classifier that does not rely on the noise. Then we will discard these conditions and introduce a more general classifier. Moreover, we will be able to adapt the classifier to the more general dynamical system described by the *stochastic differential equation (SDE)* [13] of the form

$$\begin{cases} d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + GdW(t), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (4)$$

where $W(t)$ is an m -dimensional *Brownian motion* and G is a $d \times m$ constant diffusion matrix, unique for each class. Phase trajectories $\bar{\mathbf{x}}(t)$ are *random processes* in this model. The IVP is defined analogously for the SDE, and its solution is likewise unique under some regularity conditions on f [13].

Our approach can be extended to a less direct connection between observed time series and the phase trajectories (2). Imagine the connection is given by some *observation*

map $\alpha : M \rightarrow \mathbb{R}^d$ such that

$$\mathbf{x}(t_i) = \alpha(\bar{\mathbf{x}}(t_i)) + \epsilon(t_i), \quad (5)$$

where M has dimension $k \neq d$ in general. We would like to reconstruct the $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$. The reconstruction will be based on building an *embedding* of the manifold M . In general, an embedding is a smooth map $\phi : M \rightarrow N$ into another manifold N that is a diffeomorphism onto its image $\phi(M)$. It follows that the phase trajectories $\bar{\mathbf{x}}(t) \subset M$ and $\phi(\bar{\mathbf{x}}(t)) \subset \phi(M)$ are diffeomorphic. To reconstruct the phase trajectories we need to build $\phi(\bar{\mathbf{x}}(t_i))$ from the observed time series $\mathbf{x}(t_i)$. One way to achieve this is the Whitney Embedding Theorem [14]. If α is a generic map from M to $\mathbb{R}^{2k+1}$ then α is itself an embedding, $\alpha = \phi$, and therefore $\mathbf{x}(t_i) = \phi(\bar{\mathbf{x}}(t_i))$. This recovers the original connection (2), but with the diffeomorphic phase trajectories in place of the original ones. The map is generic if it preserves information about the original dynamics. Usually this means that α is smooth and non-constant. However, this approach requires an excessive number, $2k + 1$, of time series even when k is rather small. Another approach is the Takens embedding theorem [15]. It requires only one observed scalar time series $x(t_i)$. If α is a generic map from M to $\mathbb{R}$ then the embedding is given by the map

$$\phi(\bar{\mathbf{x}}) = (\alpha(\bar{\mathbf{x}}), \alpha(f(\bar{\mathbf{x}})), \dots, \alpha(f^m(\bar{\mathbf{x}}))),$$

where $m > 2k$. Substituting $\bar{\mathbf{x}}(t_i)$ gives us $\phi(\bar{\mathbf{x}}(t_i)) = (x(t_i), x(t_{i-1}), \dots, x(t_{i-m}))$, which we call a *delay vector*. An illustration of the Takens theorem is in Fig. 1. The delay vectors form a noisy diffeomorphic phase trajectory because $x(t_i)$ is itself noisy, so they play the same role as the $\mathbf{x}(t_i)$ in (2). Henceforth we do not differentiate between reconstructed phase trajectories and the original ones.

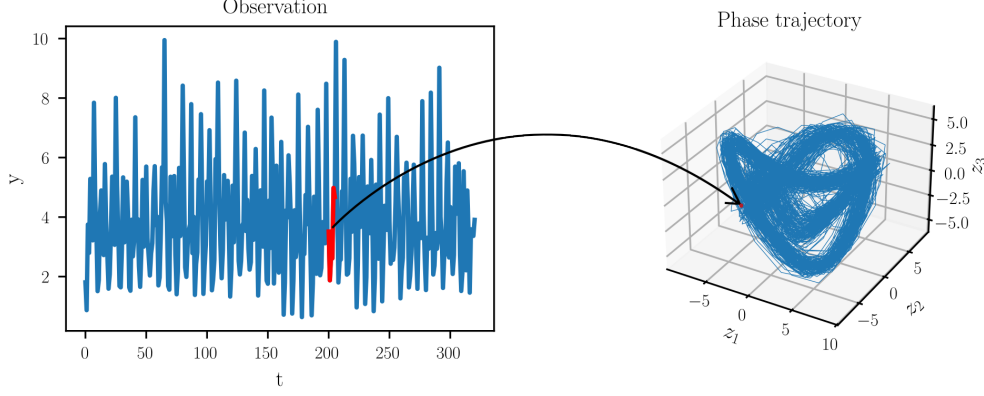


Figure 1: Application of the Takens theorem. On the left are observations $x(t_i)$. To build a point of the phase trajectory $\bar{\mathbf{x}}(t_i)$ we must take several lagged observation values (colored in red), e.g. $\bar{\mathbf{x}}(t_i) = (x(t_i), x(t_{i-1}), x(t_{i-2}))^T$.

Our time series model can be applied to many real-world classification problems. For example, in medicine an electroencephalogram (EEG) is a direct, macroscopic measure of the brain’s electrical activity [16]. The activity is governed by Maxwell’s equations, which can be reduced to ODEs [17]. The electrocardiogram (ECG) can be used for detecting arrhythmias or other heart conditions [6]. We demonstrate this further in Section 7.4. The EEG can identify seizures and sleep stages from brainwave data [7]. In human activity recognition (HAR) ODEs are primarily used to model the dynamics of human motion. The core idea is that human movement is a continuous physical process governed by the laws of mechanics, which can be described by differential equations [21]. We applied this idea to basic activity recognition in the Empirical study Section, but it is also valid for fine-grained activities [22, 23] and fall detection [24]. To summarize, our model is applicable when it is reasonable to assume that the observed time series come from some dynamical system of the form (1) or (4).

Our classifier will rely on the knowledge of the vector fields f_j , which is not always available. To compensate for this we will build approximate fields from data using Neural Ordinary Differential Equations (NODE) [25]. It allows using gradient optimization when the optimized objective depends on ODE trajectories. There are

plenty of papers using NODE for regression tasks. For example, NODE is used in the context of physics-informed ODEs [26, 27] and control learning [28, 29]. However, very few papers analyze NODE in the context of time series classification. The authors of [30] reinterpreted convolutional neural networks in terms of latent ODE dynamics and used them for hyperspectral image classification. The paper [31] solved the classification problem in terms of control theory and NODE. Our paper contributes to NODE applications for general dynamical systems.

The NODE paper [25] introduced its own popular time series model called latent ODE. It is illustrated in Fig. 2. The model is probabilistic and in our notation it can be described as

$$\begin{aligned}\bar{\mathbf{x}}_{t_0} &\sim p_0(\bar{\mathbf{x}}_{t_0}), \\ \bar{\mathbf{x}}_{t_1}, \bar{\mathbf{x}}_{t_2}, \dots, \bar{\mathbf{x}}_{t_N} &= \text{ODESolve}(\bar{\mathbf{x}}_{t_0}, f, t_0, \dots, t_N), \\ \text{each } \mathbf{x}_{t_i} &\sim p_x(\mathbf{x}_{t_i} | \bar{\mathbf{x}}_{t_i})\end{aligned}$$

where $\text{ODESolve}()$ is an approximate solution of the initial value problem using an ODE solver [32]. The authors used a variational autoencoder (VAE) [33] to infer the distributions p_0, p_x and the vector field f from the $\mathbf{x}_{t_i}$. While this model is more general and is widely used [34–36], it has several drawbacks compared to ours. First, it has to parameterize f, p_x and the VAE. Hence, more data is needed in general to obtain close approximations of these functions. Our approach may only need to parameterize f . Second, in the latent ODE model the connection between the phase trajectories and the observed time series is given by the general distribution p_x . Thus, the connection is difficult to interpret. In addition, the VAE is needed to reconstruct the $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$. Our reconstruction approaches discussed above give explicit and easy ways to obtain $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$ and vice versa. Several papers can be found

in the literature [37, 38] that use Takens theorem ideas with NODE to restore latent dynamics. Unfortunately, this is poorly covered for machine learning applications. Our paper aims to close this gap.

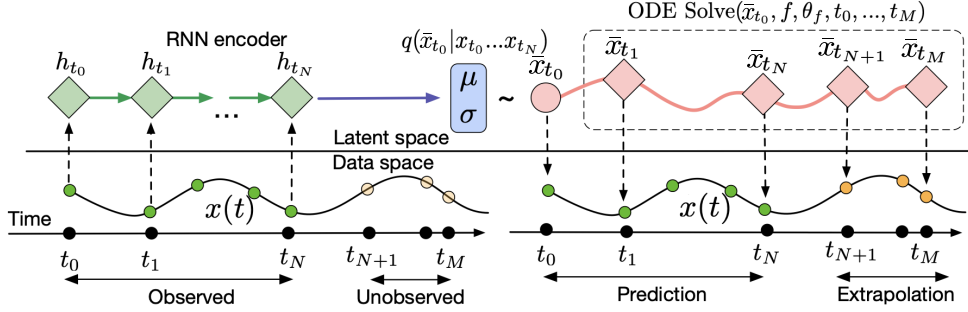


Figure 2: Computational graph of the latent ODE model. Source: adapted from Chen et al. (2018).

In the next section we will describe a general principle for the time series classification inside our model. We will build an asymptotically optimal classifier under the mentioned properties of the noise ϵ_{t_i} and several additional conditions. These conditions will help with the theoretical proof but may not hold in practice. We will then progressively discard these conditions and adapt the classifier to more general settings. We will also show how to adapt the classifier to the SDE case (4). Then we will describe the procedure of building vector field approximations from the data. Finally, we will apply our method to three classification tasks for empirical validation. The first will be the classification of three linear ODE and SDE systems. The second will be the classification of human motions from Inertial Measurement Unit (IMU) sensors. The third will be the heart failure detection from ECG univariate time series.

2 Asymptotically optimal classifier

The main principle for building a classifier in the paper is the proximity of an observed time series $\mathbf{x}(t_i)$ to the theoretical initial value problem solution $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$ for each class $j \in \{1, \dots, K\}$. Our classifier is therefore distance-based.

Denote by $\rho_n(\mathbf{x}(t_i), \mathbf{z}(t_i))$ the metric between time series $\mathbf{x}(t_i), \mathbf{z}(t_i)$ of (possibly infinite) length n defined as

$$\rho_n(\mathbf{x}(t_i), \mathbf{z}(t_i)) = \sum_{i=1}^n \|\mathbf{x}(t_i) - \mathbf{z}(t_i)\|_2^2. \quad (6)$$

Then, define the classifier as

$$\delta(\mathbf{x}(t_i)) = \arg \min_{j \in \{1, \dots, K\}} \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j)). \quad (7)$$

This theoretical classifier has a vanishing probability of misclassification as the length of the time series grows, provided the starting point $\bar{\mathbf{x}}_0$ is known exactly and the phase trajectories of different systems diverge in the sense of the statement below.

Theorem 1. *Let $\mathbf{x}(t_i) = \{\mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}\}$ be the time series to classify, defined for all $n \in \mathbb{N}$. Let $\bar{\mathbf{x}}_0$ be the starting point of the associated phase trajectory $\bar{\mathbf{x}}(t_i)$. Moreover, let*

$$\rho_\infty(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p)) = +\infty \quad \forall j \neq p; \quad j, p \in \{1, \dots, K\}. \quad (8)$$

Denote by y the true class of the time series. Then, under proposed conditions on the noise (3), the following holds

$$\mathbb{P}(\delta(\mathbf{x}(t_i)) \neq y) \rightarrow 0, \quad n \rightarrow \infty.$$

The proof of the theorem is deferred to [Appendix A](#).

3 Why optimal classifier may be intractable

Unfortunately, the classifier (7) relies on precise IVP solutions and Theorem 1 relies on a precise starting point $\bar{\mathbf{x}}_0$. These requirements are often too strong in practice, as we argue below. Then we will propose a more general classifier.

3.1 Approximate vector fields

Imagine instead of the true vector fields f_j we only have approximate versions $\tilde{f}_j$ such that

$$\|f_j - \tilde{f}_j\| \leq r_j,$$

where $f_j, \tilde{f}_j \in \mathcal{F}(M)$ and r_j are approximation error bounds. This situation always arises when one approximates vector fields from the data (see Section 6).

In general, an arbitrarily small approximation error is enough for the phase trajectories of f and $\tilde{f}$ to diverge in the (8) sense. We show such a system in Example 1 in Appendix A. The visualization of the associated trajectories is in Fig. 3. That makes classifier (7) useless. Nonetheless, for a large family of vector fields the $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}_j)$ are arbitrarily close to $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$ if r_j is sufficiently small and the times are bounded, $t < T$. These are called Grönwall-type bounds and can be found in the ODE literature [39]. So further we assume the approximation errors are small enough that we do not differentiate between $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}_j)$ and $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f_j)$. We also assume that the vector fields f_j are considerably different across classes to exclude situations like in Example 1.

3.2 Approximate IVP solutions and noisy phase trajectories

Even if the vector fields are known exactly, the IVP is almost always intractable [32]. Numerical methods known as ODE solvers approximate the IVP solution to arbitrary precision for regular vector fields [32]. The main approximation parameter of the solvers is the integration step τ . For a solver with convergence order l the general

approximation bound is

$$\sup_{t \in [0, T]} \|\text{ODESolve}(t|\bar{\mathbf{x}}_0, f) - \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)\| = \mathcal{O}(e^{LT} \tau^l), \quad (9)$$

where L is a Lipschitz constant of f . The higher the convergence order and the smaller τ , the higher the computational complexity of the ODESolve procedure. Besides this tradeoff, some vector fields imply additional constraints on the smallness of τ . If these are not satisfied, the solver solution diverges from the true one. This effect is called *stiffness* and is usually addressed by choosing an appropriate solver [40]. The illustration of this effect is in Fig. 4.

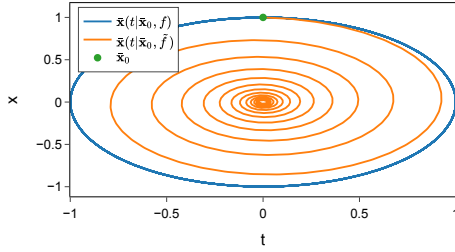


Figure 3: Diverging phase trajectories of the close vector fields $\|f - \tilde{f}\| \leq 0.1$ with $\bar{\mathbf{x}}_0 = (0, 1)^T$. While $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)$ is a circle trajectory (called a center in dynamical systems), the $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f})$ is a spiral sink that converges to $(0, 0)^T$.

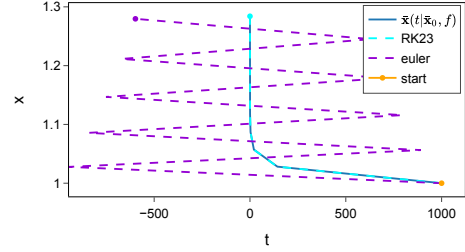


Figure 4: Exact and approximate IVP solutions for the stiff system $f(\bar{\mathbf{x}}) = \begin{pmatrix} -35 & 0 \\ 0 & 0.5 \end{pmatrix} \bar{\mathbf{x}}$ and $\bar{\mathbf{x}}_0 = (1000, 1)^T$. The Runge-Kutta23 (RK23) ODE solver approximates the exact solution with high precision. The Euler (euler) ODE solver is unstable and diverges from the exact solution. The integration step $\tau = 0.05$ is the same for both solvers, but for Euler it is not small enough. The RK23 has a higher convergence order and is stable under such τ .

As follows from (9), applying ODESolve to find the full trajectory $t \in [0, T]$ starting from the exact $\bar{\mathbf{x}}_0$ becomes impractical for long trajectories.

We also assume

$$\bar{\mathbf{x}}_0 \sim F(\cdot) \quad (10)$$

to be a random vector in general, where $F(\cdot)$ is its distribution function. Hence, the classifier (7) cannot be applied under these conditions.

On the other hand, imagine we knew $\bar{\mathbf{x}}_0$ and $\bar{\mathbf{x}}_{t_i}$ for some $i \in \{1, \dots, n\}$. We could then approximate $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)$ on $[0, t_i]$ by $\text{ODESolve}(t|\bar{\mathbf{x}}_0, f)$ and on $[t_i, T]$ by $\text{ODESolve}(t|\bar{\mathbf{x}}_{t_i}, f)$. Both integration intervals are shorter than $[0, T]$, so by (9) the solver's accuracy increases. To obtain a practical solution, we could simply replace $\bar{\mathbf{x}}_{t_i}$ with $\mathbf{x}_{t_i}$. However, if σ in (3) is large, these points differ considerably with high probability, which in general leads to very different IVP solutions $\text{ODESolve}(t|\mathbf{x}_{t_i}, f)$ and $\text{ODESolve}(t|\bar{\mathbf{x}}_{t_i}, f)$ [39]. We must take the randomness into account in a different way.

4 Classification as smoothing problem

Let us break the dynamics (1) into consecutive time segments $[t_0, t_1), [t_1, t_2), \dots, [t_{n-1}, t_n]$, so we have n IVPs

$$\begin{cases} \frac{d}{dt}\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}}), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_{t_i}, \\ t \in [0, t_{i+1} - t_i]. \end{cases} \quad (11)$$

Let us approximate each IVP solution by ODESolve, so that $\bar{\mathbf{x}}_{t_{i+1}} = \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f)$. Combining this with (2) and (10), we obtain the discrete dynamics

$$\begin{cases} \bar{\mathbf{x}}_{t_{i+1}} = \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f), \\ \mathbf{x}_{t_i} = \bar{\mathbf{x}}_{t_i} + \epsilon_{t_i}, \\ \bar{\mathbf{x}}_0 \sim F(\cdot). \end{cases} \quad (12)$$

Only $\mathbf{x}(t_i)$ is observed; everything else is latent. To adapt classifier (7) to this model we need to replace the intractable $\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f)$ with its best approximation given the random $\bar{\mathbf{x}}_0$ and $\mathbf{x}_{t_i}$, $i \in \{1, \dots, n\}$. Under the L_2 norm in the space of random vectors, this best approximation is the *conditional expectation* [41]. We denote the series of conditional expectations by $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$, meaning that its i -th value is $\mathbb{E}[\bar{\mathbf{x}}_{t_i} | \bar{\mathbf{x}}_0, \mathbf{x}_{t_1}, \dots, \mathbf{x}_{t_n}]$. Finding $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ is called the *smoothing problem*. Like the IVP, it is intractable in general but admits numerical approximation. Some methods for that are extensions of the Kalman filter [42, 43]; others are based on Bayesian and particle filters [44].

The new classifier is defined as

$$\delta(\mathbf{x}(t_i)) = \arg \min_{j \in \{1, \dots, K\}} \rho_n(\mathbf{x}(t_i), \mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f_j]). \quad (13)$$

The classifier now relies not only on the dynamics driven by the vector fields f_j but also on the observations $\mathbf{x}(t_i)$. This mitigates the problems discussed in Section 3:

1. Inaccuracy from the approximate $\tilde{f}$ and ODESolve decreases because we give up the full IVP $t \in [0, T]$ in favor of the smaller IVPs on time segments $t \in [t_i, t_{i+1}]$.
2. Randomness of the $\bar{\mathbf{x}}_0, \mathbf{x}_{t_i}$ is taken into account via the conditional expectation $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. Informally, it optimally balances the underlying dynamics (1) with the noisy observations to approximate the IVP solution $\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f)$.

Comparison of the IVP $\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f)$ approximation by $\text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f)$ and by $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ is illustrated in Fig. 5.

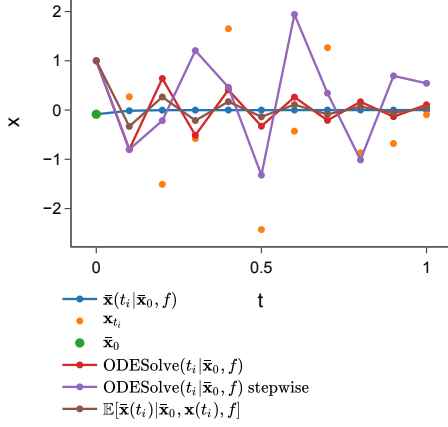


Figure 5: Different IVP approximations for the one-dimensional system $f(x) = -18x$. The initial value $x_0 \sim \mathcal{N}(1, 1)$ and $\epsilon_{t_i} \sim \mathcal{N}(0, 1)$. ODESolve uses Euler solver with initial value $\mathbb{E}[x_0] = 1$. Stepwise ODESolve uses the Euler solver on consecutive time segments $[t_i, t_{i+1})$ with noisy initial values x_{t_i} . The Unscented Kalman Filter (UKF) [43] is used to compute $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. UKF uses stepwise Euler internally. We can see that UKF demonstrates the best IVP approximation by using both the underlying dynamics and the noisy observations. UKF is the most accurate IVP approximation with probability 0.92 by Monte-Carlo estimation.

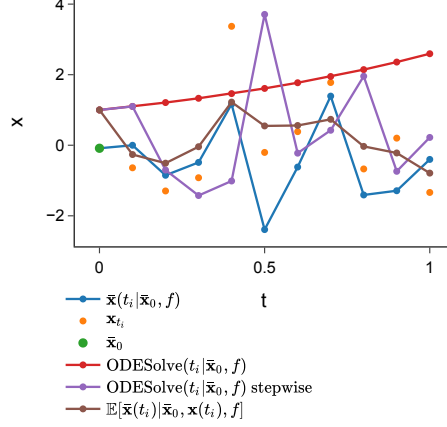


Figure 6: Different IVP approximations for the one-dimensional system $dx(t) = xdt + dW(t)$. The initial value $x_0 \sim \mathcal{N}(1, 1)$ and $\epsilon_{t_i} \sim \mathcal{N}(0, 1)$. The phase trajectory is a random process here, in contrast to the ODE dynamics. ODESolve uses the Euler solver with initial value $\mathbb{E}[x_0] = 1$. We can see it is an inadequate model for the SDE dynamics. Stepwise ODESolve uses the Euler solver on consecutive time segments $[t_i, t_{i+1})$ with noisy initial values x_{t_i} . It is more accurate but is subject to the noise and still misses the $dW(t)$ term. The Unscented Kalman Filter (UKF) is used to compute $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. UKF uses stepwise Euler internally but also accounts for the $dW(t)$ term. It is the most accurate IVP approximation among those mentioned with probability 0.97 by Monte-Carlo estimation.

5 Adapting classifier for SDE

We can segment the dynamics (4) similarly to (11) and obtain

$$\begin{cases} d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + GdW(t), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_{t_i}, \\ t \in [0, t_{i+1} - t_i]. \end{cases} \quad (14)$$

If all time segments are small enough, we can approximate the dynamics $f(\bar{\mathbf{x}})dt \approx \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f)$ and $GdW(t) \approx G \cdot \mathcal{N}(0, (t_{i+1} - t_i) \cdot I) = \mathcal{N}(0, (t_{i+1} - t_i) \cdot GG^T) := \xi_{t_{i+1}}$. As a result, we obtain discrete dynamics akin to (12):

$$\begin{cases} \bar{\mathbf{x}}_{t_{i+1}} = \text{ODESolve}(t_{i+1} - t_i | \bar{\mathbf{x}}_{t_i}, f) + \xi_{t_{i+1}}, \\ \mathbf{x}_{t_i} = \bar{\mathbf{x}}_{t_i} + \epsilon_{t_i}, \\ \bar{\mathbf{x}}_0 \sim F(\cdot). \end{cases} \quad (15)$$

Finding $\mathbb{E}[\bar{\mathbf{x}}(t_i) | \bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$ here is just a more general smoothing problem and can be solved by the same methods listed above. The classifier (13) for the SDE case stays the same. Comparison of the different SDE IVP $\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f)$ approximations is given in Fig. 6.

6 Learning vector field from data

Sometimes the vector field f may be unknown. Instead, time series samples from the associated dynamics (1) may be available. From (1) and (2) we have

$$\mathbf{x}_{t_i} = \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f) + \epsilon_{t_i}, \quad i \in \{1, \dots, n\},$$

where $\bar{\mathbf{x}}_0$ is defined as in (10). Approximating the true IVP by ODESolve, we obtain

$$\mathbf{x}_{t_i} = \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f) + \epsilon_{t_i}, \quad i \in \{1, \dots, n\}.$$

By taking the expectation of both sides and gathering the terms on the left-hand side, we get

$$\mathbb{E}[\mathbf{x}_{t_i} - \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f)] = 0, \quad i \in \{1, \dots, n\}.$$

We can apply the *generalized method of moments* (GMM) [45] here. Assume we have a set of vector fields $\mathcal{S}_\Theta \subset \mathcal{F}(M)$ parameterized by the real vector Θ . Assume $f \in \mathcal{S}_\Theta$. Denote the solution of the following optimization problem

$$\sum_{i=1}^n \|\mathbf{x}_{t_i} - \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f)\|_2^2 \rightarrow \min_{f \in \mathcal{S}_\Theta} \quad (16)$$

as $f_{\hat{\Theta}}$. Here $\bar{\mathbf{x}}_0, \mathbf{x}_{t_i}, i \in \{1, \dots, n\}$, are sampled values. If some technical conditions on Θ hold [45], the GMM guarantees that

$$f_{\hat{\Theta}} \xrightarrow{\mathbb{P}} f, \quad n \rightarrow \infty. \quad (17)$$

Problem (16) can be solved using NODE. As discussed in Section 3, n should not be large due to approximation errors of the ODESolve. Instead, it is better to sample N short batches $B_j = (\bar{\mathbf{x}}_0, \{\mathbf{x}_{t_i}\}_{i=1}^n)$ and solve

$$\sum_{j=1}^N \sum_{\bar{\mathbf{x}}_0, \mathbf{x}_{t_i} \in B_j} \|\mathbf{x}_{t_i} - \text{ODESolve}(t_i|\bar{\mathbf{x}}_0, f)\|_2^2 \rightarrow \min_{f \in \mathcal{S}_\Theta}, \quad (18)$$

which is equivalent to (16) in terms of the convergence (17).

SDE dynamics (4) can also be learned from data by methods similar to NODE [46, 47].

7 Empirical study

In this section we validate the proposed classifier (13) on three classification tasks with progressively decreasing prior knowledge about the underlying dynamics. In the first task the vector fields f_j are known analytically, which lets us isolate the behavior of the smoothing-based classifier from the error of the field approximation. In the second task the fields are unknown and are learned from data with NODE as described in Section 6. The phase trajectories are still observed directly through a multivariate IMU sensor. Data is taken from the [BasicMotions](#) dataset [48]. In the third task we only have univariate time series. Before field learning, we will apply Takens theorem to obtain the phase trajectories. Data is taken from the [ECG5000](#) dataset [48].

7.1 Experimental setup

Across all tasks we evaluate the classifier (13) where the conditional expectation $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f_j]$ is approximated with the Unscented Kalman Filter together with the Rauch–Tung–Striebel smoother [44].

Classification quality is reported as accuracy on a held-out test set. All trajectory integration and vector field learning are performed with `torchdiffeq`, and the trajectory smoothing with `filterpy`.

7.2 Linear dynamical systems

The first task classifies trajectories generated by three linear ODE systems with the following form

$$f(\bar{\mathbf{x}}) = A\bar{\mathbf{x}} = U\Lambda U^T\bar{\mathbf{x}} = U \begin{pmatrix} -\gamma_1 & w & 0 \\ -w & -\gamma_1 & 0 \\ 0 & 0 & -\gamma_2 \end{pmatrix} U^T\bar{\mathbf{x}}. \quad (19)$$

Here A is a real matrix with the specified block decomposition, where U is orthogonal. All three systems share the same Λ but differ in the matrix U . The vector field (19) simulates damped oscillation along the first two basis directions and exponential decay along the third. The basis directions are the U columns. We used $\gamma_1 = 0.7, \gamma_2 = 0.6, w = 4$. For each class we sample $\bar{\mathbf{x}}_0 \sim \mathcal{N}(\mathbf{0}, 3^2 \cdot I)$ and integrate the system trajectories of length $n = 100$ with step $\tau = 0.05$. Then we corrupt them with additive Gaussian noise of different variance levels σ^2 following (2)–(3). The visualization of the final series and the underlying phase trajectories is in Fig. 7.

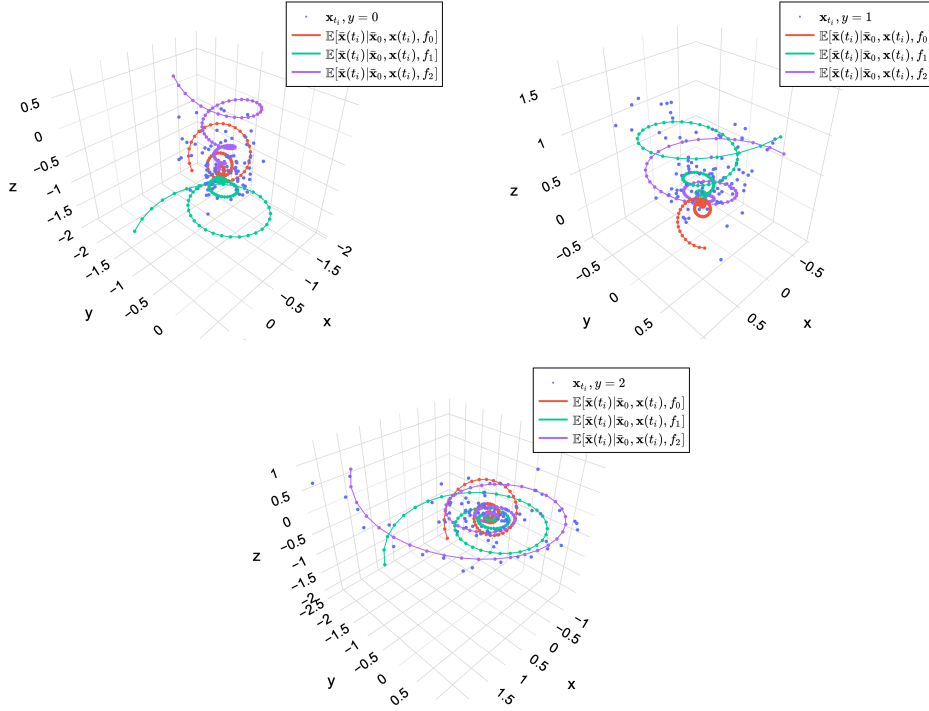


Figure 7: Observed time series $\mathbf{x}(t_i)$ generated from each linear ODE system (y label in the figure legend) and smoothed phase trajectories from every field f_0, f_1, f_2 . The systems are a combination of exponential decay and damped oscillations. Noise level for the illustration is $\sigma = 0.2$.

Because the true fields f_j are available, no field learning is required. We compare our classifier against time series classification baselines: a k -Nearest-Neighbors classifier with the Dynamic Time Warping distance [2] and a recurrent (LSTM) network trained end-to-end on the raw series. Both baselines are relevant to our model. Recurrent networks have internal discrete dynamics that can be viewed as Euler solver steps [25]. DTW defines a distance between time series [2] that differs from (6).

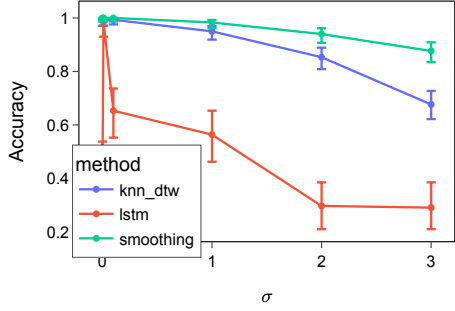


Figure 8: Linear systems’ ODE classification accuracy with different noise levels σ of the observations. Error bars correspond to the accuracy’s confidence intervals calculated via Wilson method.

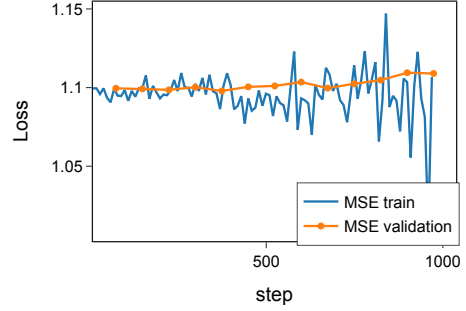


Figure 9: Learning curves for the LSTM training with $\sigma = 2$: MSE loss on train and validation series.

Classification accuracy and its confidence intervals for the three methods are reported in Fig. 8. LSTM shows inferior performance compared to the distance-based classifiers. The model struggles to differentiate the linear systems due to their “proximity” implied by the shared Λ . It is completely unable to learn ODE dynamics for high noise levels, as also seen from the training curves in Fig. 9. The k -NN + DTW classifier is distance-based like (13) but is sub-optimal. It uses DTW to compute the series distance instead of (6) and $\mathbb{E}[\bar{\mathbf{x}}(t_i)|\bar{\mathbf{x}}_0, \mathbf{x}(t_i), f]$. It also learns f_j only indirectly by memorizing the training series for each class. Classifier (13) combines information

from both f_j and the noise level and shows significantly better performance. The monotonic decrease in accuracy is expected, given the higher noise level and the limited trajectory length.

We also transformed the previous ODE systems into SDEs of the form

$$d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + \sigma_W I \cdot dW(t),$$

where f is the same as in (19), $\sigma_W = 0.1$. We conducted the same classification experiment varying σ as in the ODE case. The visualization of the SDE time series and the underlying phase trajectories is in Fig. 10. Classification accuracy is reported in Fig. 11 and has the same behavior as in the ODE case.

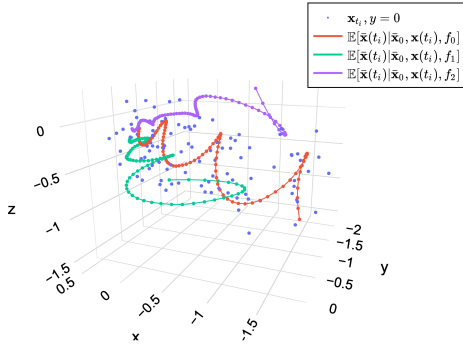


Figure 10: Observed time series $\mathbf{x}(t_i)$ generated from SDE system with $y = 0$ and smoothed phase trajectories from every field f_0, f_1, f_2 . The vector fields are the same as in the ODE case. Noise level for the illustration is $\sigma = 0.2$.

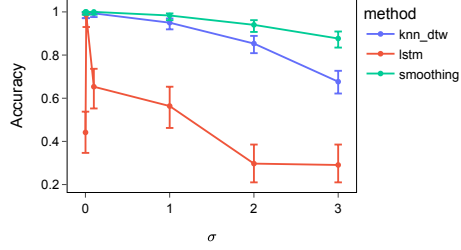


Figure 11: Linear systems' SDE classification accuracy with different noise levels σ of the observations. Error bars correspond to the accuracy's confidence intervals calculated via Wilson method.

7.3 Human motion recognition from IMU

The second task uses the BasicMotions dataset of Inertial Measurement Unit (IMU) recordings. The dataset contains the rotation rate $(w_x, w_y, w_z)^T$ and acceleration $(a_x, a_y, a_z)^T$ from the IMU (located on the wrist) sampled at 10 Hz. Subjects performed

4 activities: standing, walking, running, and badminton, repeating each 5 times. Every activity corresponds to a classification class. The train and test splits each contain 2 subjects, which makes 40 six-dimensional time series per split. The series length is 100 measurements.

The human body in motion is well approximated as a system of articulated rigid bodies (limbs and trunk coupled at joints), and the motion of such a system is governed by the Lagrangian equations of mechanics [49, 50]. These equations can be written in the form (1). The IMU measures only part of the full system, but we show below that this suffices for building accurate classifier. Steady locomotion activities (walking, running, badminton) are rhythmic and quasi-periodic, which in dynamical-systems language corresponds to stable limit cycles. Static postures (standing) correspond to stable equilibria.

We apply both the ODE and SDE approaches for this task. Unlike the previous experiment, the vector fields/diffusion terms for the activities are unknown. In the ODE case, for each class we learn vector field approximations from short trajectory batches by solving the objective (18). In the SDE case, we discretized the dynamics (4) with the Euler-Maruyama scheme [51], which implies the Gaussian transition

$$\bar{\mathbf{x}}_{t_{i+1}}|\bar{\mathbf{x}}_{t_i} \sim N(\bar{\mathbf{x}}_{t_i} + f(\bar{\mathbf{x}}_{t_i})(t_{i+1} - t_i), (t_{i+1} - t_i) \cdot GG^T).$$

The fields f_j and the diffusion matrices G_j are then learned by maximizing the likelihood of the train trajectories. In both approaches, the vector fields were parameterized by neural networks.

We adopt (2) as the model for the IMU time series. We take the rotation rate as the phase trajectory $\mathbf{x}(t)$ because its derivative is directly connected to rigid body mechanics. Acceleration is itself a derivative of the velocity and should not be included in the ODE/SDE dynamics. However, it indirectly contains properties of the motion,

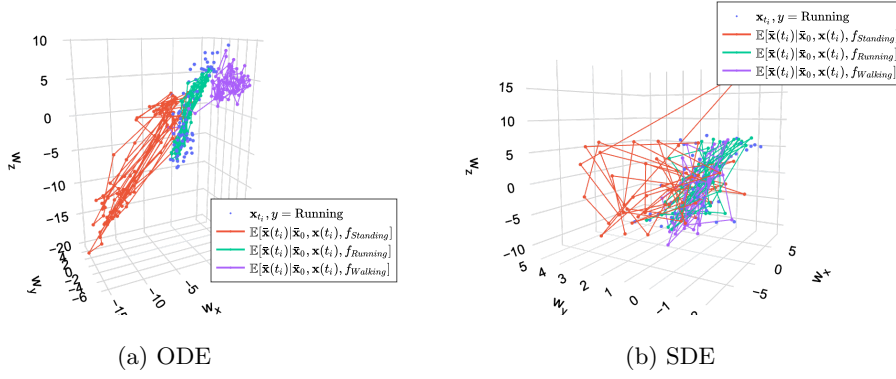


Figure 12: Rotation rate time series and their smoothing using learned vector fields from different activities. The time series are from the “running” class. The smoothing is from the “running”, “walking”, and “standing” classes.

for example, speed when running or walking. So we embed acceleration into the dynamics as a *control* variable — we let the vector field and diffusion matrices depend on it. For example, the ODE dynamics is now written as

$$\frac{d}{dt}(\bar{w}_x(t), \bar{w}_y(t), \bar{w}_z(t))^T = f(\bar{w}_x(t), \bar{w}_y(t), \bar{w}_z(t), \bar{a}_x(t), \bar{a}_y(t), \bar{a}_z(t))$$

One technical caveat: ODE solvers must evaluate f at any $t \in [0, t_n]$, but acceleration is known only at the discrete times t_i . To overcome this, we cubically interpolate the acceleration series. This control variable idea resembles Neural Controlled Differential Equations [52], where series values also control the ODE dynamics but in a slightly different way. We also conduct the experiment without acceleration as a control to prove the idea’s utility.

We empirically chose the noise covariance matrix to be $R = \sigma^2 I$, where $\sigma = 1$ for the ODE and $\sigma = 10^{-3}$ for the SDE. Smoothed trajectories were stable and stayed close to the observed trajectories under this choice. When only approximate fields are available, we have not found a more rigorous and systematic way to estimate R .

Visualization of the rotation rate time series and smoothed trajectories corresponding to different activities is in Fig. 12. Smoothing in the SDE case is more concentrated around the $\mathbf{x}_{t_i}$ due to the small σ — the smoother’s penalty for deviation from the observation is roughly inversely proportional to σ . We can see that the smoothing produced by the field of the same class as the time series stays closer to that series than the smoothing from the other classes. It is less evident in the SDE case, but computing the ρ_n distances confirms it.

Activity classification accuracy is reported in Table 1. The ODE approach correctly classified all test trajectories and showed the best result. We can see that adding control variables was necessary; without them, smoothing fails (column 2). For the SDE approach, control variables were not as important. We suppose that the diffusion term GdW_t alone was enough to encapsulate rotation rate variations caused by unseen speed variations. Moreover, the SDE approach was not able to properly learn badminton dynamics from the data and therefore showed worse performance.

Table 1: Activity classification accuracy on BasicMotions IMU data.

	ODE smooth	ODE smooth (no controls)	SDE smooth	SDE smooth (no controls)
running	1	0	1	1
walking	1	1	1	1
standing	1	0.5	0.9	0.8
badminton	1	0	0.1	0.4
all	1	0.375	0.75	0.8

7.4 Heart failure detection from ECG

The third task uses the ECG5000 dataset containing univariate electrocardiogram recordings 140 samples long each. The heartbeats were extracted from a single patient with severe congestive heart failure and divided into five beat types: one normal rhythm (class 1) and four kinds of abnormal beats (classes 2-5). Sample recordings for every class are shown in Fig. 13. The class frequencies are highly imbalanced, and the

train size for classes 3-5 was too small to learn proper vector fields. For that reason we restricted ourselves to classes 1 and 2 only.

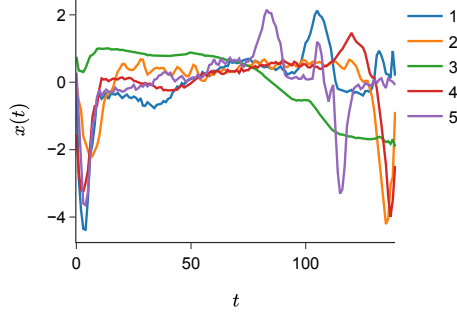


Figure 13: Sample time series for each class in the ECG5000 dataset.

Unlike the previous tasks, here we observe only a scalar signal $x(t_i)$ instead of the full phase trajectory. We therefore reconstruct the trajectory with the Takens delay embedding (5). Stacking $d = 20$ consecutive lagged samples turns each 140-point recording into a trajectory $\bar{\mathbf{x}}(t_i)$ of 131 points in the 20-dimensional phase space. The per-class vector fields f_j are unknown and are learned with NODE from short trajectory batches by solving the objective (18). The noise covariance matrix was empirically set as $R = \sigma^2 I$, where $\sigma = 10^{-3}$.

As a result, we achieved 0.970 accuracy for $y = 1$, 0.954 for $y = 2$. This validates the Takens approach for univariate time series and shows the same performance on the restricted classes as feature-based and deep learning baselines [53]. However, the approach has a downside: we could not learn proper vector fields in the high-dimensional space with a small sample size for the other classes.

8 Conclusion

In the paper we have introduced a distance-based time series classifier. It assumes a particular time series model — ODE or SDE dynamical systems. We proved it

is asymptotically accurate under idealized conditions and reasonably adapted it to the more general case. Our classifier is simple and powerful as it requires only the systems' vector fields f_j . It then uses the well-developed framework of ODE solvers and smoothing methods. Our classifier is also flexible. If we do not know the vector fields, we can learn them from data. If we do not observe the entire dynamics, we can use the Whitney/Takens theorems to restore it. The classifier's flexibility and applicability were demonstrated empirically on human motion signals and ECG recordings.

Our classifier has certain limitations by design. Even though the introduced dynamical systems are quite general, they cannot describe any time series. For example, we expect poor performance for non-continuous dynamics or general random processes. Next, learning vector fields may be cumbersome if we have no clue about their form or lack data. The same applies to the noise covariance estimation. Finally, the numerical smoothing methods the classifier uses have their own conditions and bounds that should be understood before application.

There are several directions to extend the current research. First, the classifier may be improved if the exact distributions of the noise and the initial value are known (e.g. Student or Uniform). The theoretical classifier could be derived and empirically validated. Second, the classifier may be adapted to more general SDE dynamics. Currently, the diffusion term is $GdW(t)$ because most smoothing methods require i.i.d. diffusion samples. The general term $g(x(t), t)dW(t)$ produces dependent samples with different distributions. Lastly, the classifier may be adapted to the *change point detection* task. The change point would correspond to shifting from one dynamical system to the other. However, this requires separate investigation.

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Appendix A Proofs and examples

Proof of Theorem 1. First, estimate the conditional probability of misclassification

$$\begin{aligned} \mathbb{P}(\delta(\mathbf{x}(t_i)) = j | y = p) &= \mathbb{P}\left[\bigcap_{s=1, s \neq j}^K \{\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_s))\} \mid y = p\right] \leq \\ &| \text{monotonicity of probability} | \leq \mathbb{P}[\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)) \mid y = p]. \end{aligned}$$

Expand the metrics in the last event using $y = p$:

$$\begin{aligned} \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)) &= \sum_{i=1}^n \|\mathbf{x}(t_i) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)\|^2 = \sum_{i=1}^n \|\boldsymbol{\epsilon}_{t_i}\|^2, \\ \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) &= \sum_{i=1}^n \|\mathbf{x}(t_i) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)\|^2 = \sum_{i=1}^n \|(\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) + \boldsymbol{\epsilon}_{t_i}\|^2 = \\ &= \sum_{i=1}^n \|\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)\|^2 + 2 \sum_{i=1}^n \langle \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle + \sum_{i=1}^n \|\boldsymbol{\epsilon}_{t_i}\|^2. \end{aligned}$$

Here $\langle \cdot, \cdot \rangle$ is the standard scalar product. The last event is now

$$\begin{aligned} \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)) &\Leftrightarrow \\ \sum_{i=1}^n \langle \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle < -\frac{1}{2} \sum_{i=1}^n \|\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)\|^2 &\Leftrightarrow \\ \sum_{i=1}^n \langle \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle < -\frac{1}{2} \rho_n(\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)). \end{aligned}$$

Apply Chebyshev's inequality and use (3):

$$\mathbb{P}[\rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j)) < \rho_n(\mathbf{x}(t_i), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p)) \mid y = p] \leq 4 \frac{\mathbb{D}\left[\sum_{i=1}^n \langle \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle\right]}{\rho_n^2(\bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i | \bar{\mathbf{x}}_0, f_p))} =$$

$$\begin{aligned}
&= 4 \frac{\sum_{i=1}^n \mathbb{E}[\langle \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \boldsymbol{\epsilon}_{t_i} \rangle^2]}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n (\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j))^T \mathbb{E}[\boldsymbol{\epsilon}_{t_i} \boldsymbol{\epsilon}_{t_i}^T] (\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j))}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n (\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j))^T R(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j))}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4 \frac{\sum_{i=1}^n \langle \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), R(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j)) \rangle}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} \leq \\
&\leq 4\sigma^2 \frac{\sum_{i=1}^n \|\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j)\|^2}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} = \\
&= 4\sigma^2 \frac{\rho_n(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))}{\rho_n^2(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} = \frac{4\sigma^2}{\rho_n(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p))} := \xi_n.
\end{aligned}$$

By the theorem condition we have

$$\rho_n(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p)) \rightarrow \rho_\infty(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_j), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f_p)) = +\infty, \quad n \rightarrow \infty,$$

so $\xi_n \rightarrow 0$, $n \rightarrow \infty$ for all $j, p \in \{1, \dots, K\}$.

Second, rewrite the probability of misclassification via conditional probabilities:

$$\begin{aligned}
\mathbb{P}(\delta(\mathbf{x}(t_i)) \neq y) &= \sum_{p=1}^K \mathbb{P}(y = p) \mathbb{P}(\delta(\mathbf{x}(t_i)) \neq p | y = p) \leq \sum_{p=1}^K \mathbb{P}\left(\bigcup_{j=1, j \neq p}^K \{\delta(\mathbf{x}(t_i)) = j\} \mid y = p\right) \leq \\
| \text{union bound} | &\leq \sum_{p=1}^K \sum_{j=1, j \neq p}^K \mathbb{P}(\delta(\mathbf{x}(t_i)) = j | y = p) \leq \sum_{p=1}^K \sum_{j=1, j \neq p}^K \xi_n = K(K-1)\xi_n \rightarrow 0.
\end{aligned}$$

□

Example 1. Here we construct a system of two arbitrarily close vector fields with diverging phase trajectories. Let $M = \{\bar{\mathbf{x}} \in \mathbb{R}^2 \mid \|\bar{\mathbf{x}}\|_2 \leq R\}$ be a disk in the plane.

Define the vector field's norm as the sup norm

$$\|f\| = \sup_{\bar{\mathbf{x}} \in M} \|f(\bar{\mathbf{x}})\|_2.$$

Let $\bar{\mathbf{x}}_0 = c \cdot (\sin \gamma, \cos \gamma)^T \neq \mathbf{0}$, $c > 0$, be an initial value. Define the vector fields as

$$\begin{aligned} f(\bar{\mathbf{x}}) &= \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix} \bar{\mathbf{x}} = A\bar{\mathbf{x}}, \\ \tilde{f}(\bar{\mathbf{x}}) &= \begin{pmatrix} -b & -a \\ a & -b \end{pmatrix} \bar{\mathbf{x}} = \tilde{A}\bar{\mathbf{x}}, \end{aligned}$$

where $a \neq 0, b > 0$ are some scalars. Then

$$\|f - \tilde{f}\| = \sup_{\bar{\mathbf{x}} \in M} \|(A - \tilde{A})\bar{\mathbf{x}}\|_2 \leq \|A - \tilde{A}\|_2 \cdot R = \left\| \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix} \right\|_2 \cdot R = bR,$$

where $\|A - \tilde{A}\|_2$ is the matrix spectral norm. So if $b < \frac{r}{R}$ then $\|f - \tilde{f}\| < r$. Next, it is easy to obtain that the IVP solutions for the ODE systems are

$$\begin{aligned} \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f) &= c \begin{pmatrix} \sin(\gamma - at) \\ \cos(\gamma - at) \end{pmatrix}, \\ \bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}) &= c \exp(-bt) \begin{pmatrix} \sin(\gamma - at) \\ \cos(\gamma - at) \end{pmatrix}. \end{aligned}$$

It is obvious that $\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, \tilde{f}) \rightarrow \mathbf{0}, t \rightarrow \infty$ and $\|\bar{\mathbf{x}}(t|\bar{\mathbf{x}}_0, f)\| = \|\bar{\mathbf{x}}_0\| = c \forall t$. So

$$\|\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})\| \rightarrow c \neq 0, \quad i \rightarrow \infty,$$

and then

$$\rho_n(\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f), \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})) = \sum_{i=1}^n \|\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, f) - \bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})\|_2^2 \rightarrow \infty, \quad n \rightarrow \infty,$$

because the necessary condition for series convergence does not hold.

As a result, the classifier (7) that uses $\bar{\mathbf{x}}(t_i|\bar{\mathbf{x}}_0, \tilde{f})$ does not have the guarantees of Theorem 1 even though f and $\tilde{f}$ are arbitrarily close.